

VINAYAK GUPTA

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EDUCATION

Birla Institute of Technology and Science, Pilani

MSc (Hons) in Biological Sciences | BE (Hons) in Manufacturing Engineering

2022 – Present

EXPERIENCE

Research Intern - Nanyang Technological University

Aug '26 – Present

Advisor: Assistant Professor Xueming Dong

Singapore

Hydrogen-Deuterium Exchange Epitope-omics Study on Intact Dengue Virus

- Analyzed HDX-MS datasets for DENV2 capsid(C), envelope(E), and membrane(M) proteins using HDX Workbench and HX-Express to quantify peptide-level deuterium uptake and antibody-induced protection.
- Performed mammalian cell culture using Baby Hamster Kidney (BHK-21) cells, including passaging, seeding, fixation, staining, and plaque assays for DENV2 propagation and infection experiments.

Research Intern - CSIR-Institute of Genomics and Integrative Biology

Feb '26 – Jul '26

Advisor: Professor Lipi Thukral

Delhi, India

Molecular Dynamics Simulations for RNA-Lipid Interactions

- Designed and executed molecular dynamics simulations of RNA molecules using GROMACS, including system preparation, solvation, equilibration, production simulations, and trajectory analysis.
- Investigated RNA structural stability and conformational dynamics through trajectory analysis and interpretation of simulation-derived structural parameters, such as RMSD, RMSF, radius of gyration.

Reproduction of the CplxCavity RNA Binding Site Prediction Framework

White Paper

- Reproduced and extended the CplxCavity RNA-small-molecule binding-site prediction framework using MDAnalysis, FreeSASA, geometric cavity detection, and physicochemical descriptors.
- Developed logistic-regression and tree-based classification models for RNA ligand-binding cavity prediction, performed feature selection and evaluated performance using MCC and ROC-AUC metrics.

PUBLICATION AND POSTER

Acoustic-Assisted HDX-MS for Intact Virion Analysis. ChemRxiv 2026 (0929)

Paper

Pang, L. C.; Deng, Y.; Vinayak Gupta; Lo, W. Z.; Zhu, S.; Shan, M.; Zhang, M., Mei; Dong, Xueming

A Scalable HDX-MS Epitope-omics Workflow for Intact Dengue Virus at 90% Sequence Coverage for Structural Proteins. HUPO 2026

Poster

Pang, L. C.; Deng, Y.; Vinayak Gupta; Lo, W. Z.; Zhu, S.; Shan, M.; Zhang, M., Mei; Dong, Xueming

PROJECTS

Antimicrobial Resistance(AMR) Prediction for Whole Genome Sequences using ML

Advisor: Professor Sandhya Marathe & Professor Tanmaya Mahapatra

Aug '25 – Dec '25

- Curated AMR datasets across 700+ drug-pathogen pairs for machine learning model development.
- Integrated genomic features and MIC values to improve accuracy and generalization in prediction models.

CNN-Based Model for Pneumonia Classification Using Chest CT Data

Dec '24

- Built a deep learning classification model to differentiate between healthy and pneumonia-affected lungs.
- Implemented image preprocessing pipelines using OpenCV and developed models in PyTorch for training.

Bioinformatics Course Project

Advisor: Professor Shibasish Chowdhury

Jan '25 – May '25

- Predicted the structural (introns, exons, Promoter regions) and coding regions using GENSCAN.
- Created phylogenetic trees of six mammalian species using MEGA11, to find maximum parsimony.
- Pairwise gene alignment of multiple gene sequences using Basic Local Alignment Search Tool(BLASTn).

Independent Study: Beta-Glucans and their Applications in Human Health and Plants

Advisor: Professor Balakumaran Chandrasekar

Jan '24 – May '24

- Conducted a literature review on the biological and agricultural applications of beta-glucans.
- Explored potential use cases in immunomodulation, disease prevention and crop protection.

Applied Biology Project Lab

Advisor: Professor Shilpi Garg

Jan '24 – May '24

- Conducted a series of experiments to culture microorganisms, such as *E. coli*, to learn wet lab practices.
- Conducted assays including MICP, MIC, Beta-galactosidase activity and MTT cell viability.

SKILLS & TECHNICAL PROFICIENCY

Languages: Python [Fast.ai, Numpy, Pandas, Biopython, Scikit-Learn], C++, C, Git, HTML, bash

Bioinformatics: PyMol, Transcriptomic Data Analysis, HDX Workbench, GROMACS, ChimeraX

Wet Lab: Mammalian cell culture of Baby Hamster Kidney cell line(BHK-21)(plaque assay, passaging, fixation & staining, infecting), MIC cell viability assay, MTT and MICP assays, Beta-galactosidase assay.

Design & Simulation: AutoCAD, Fusion 360, Autodesk Maya, Arduino

COURSEWORK

Computational Biology: Biomolecular modeling, Bioinformatics, Recombinant DNA Technologies, Computer programming, Probability & Statistics, Machine Drawing, Engineering graphics

Biological Sciences: Developmental Biology, Immunology, Biological Chemistry, Genetics, Biophysics, Microbiology, Ecology & Environmental Sciences, Animal & Plant Physiology, Cell Biology